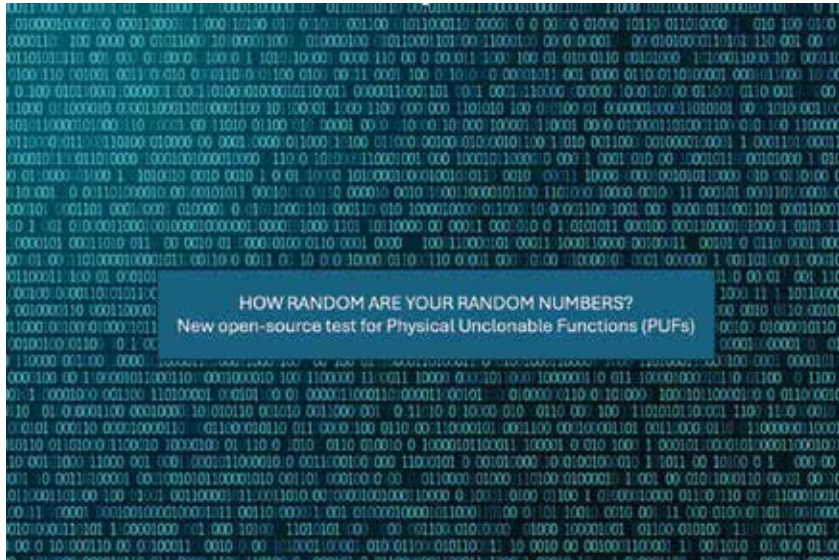


TuRiNG: Test suite for enhancing PUF security



Crypto Quantique has unveiled a randomness test suite designed specifically for physical unclonable functions (PUFs). This new suite, named TuRiNG, addresses the unique challenges of evaluating PUFs, which are increasingly integral to semiconductors for secure and regenerable random number generation—essential for ensuring device security, creating unique device identities, and generating cryptographic keys. PUFs differ from traditional random number generators (RNGs) in that they produce fixed-length outputs, rendering existing randomness tests inadequate. Ensuring randomness is crucial for maintaining a secure device root-of-trust (RoT), and TuRiNG has been specifically developed to meet this need.

The suite adapts tests from the widely recognised NIST 800-22 randomness test suite, tailoring them to the distinct characteristics of PUFs, including their spatial dependencies and limited output lengths. A key feature is a test designed to ensure the independence of PUF outputs. This is vital for upholding cryptographic security, as it helps identify and mitigate the risk of correlated outputs, which could otherwise compromise the security of cryptographic systems. The suite is both comprehensive and practical, balancing thoroughness with efficiency to ensure PUFs meet randomness requirements without demanding excessive amounts of data.

TuRiNG also introduces innovative techniques, such as running tests in different data orderings to account for potential spatial correlations in PUF outputs. By reducing the number of required bits for certain tests, the suite enables more efficient testing, minimising the risk of misrepresenting PUF quality and enhancing the overall reliability of the randomness validation process.

SambaNova's new AI chip takes on NVIDIA's dominance in the chip market

SambaNova Systems has introduced a cloud service for AI software developers, positioning itself among competitors like Cerebras and Groq to challenge NVIDIA's dominance in the AI chip market. The company launched SambaNova Cloud, a new

GNOME 47 released

GNOME 47 has been released as the latest major update to the popular open source desktop environment.

Key features of GNOME 47 (codename Denver) include support for nine accent colours in the 'Style' section of the 'Appearance' panel in 'Settings'. It also introduces the 'File Chooser' portal, enabling larger thumbnails when opening files in apps, and the implementation of the XDG dialog protocol on Wayland to specify hints on surfaces like dialog modality.

Notable improvements include support for the DRM lease protocol, the reintroduction of support for legacy X11 cursor themes, three-finger touchpad gestures, persistent remote login sessions, and hardware encoding for screen recordings. Additionally, it adds support for tablet tool key bindings and pressure ranges.

GNOME 47 brings experimental support for colour management protocols in the 'Mutter' window and composite manager, initial PipeWire explicit sync support, logic improvements for placing and focusing always-on-top windows, and ensures the notification portal always uses the GTK protocol. It also supports building GNOME Shell without X11, aligning with a Wayland-only future.

Enhancements include better lock screen notifications, centred time display when using a 12-hour format, improved monitor matching for window restoration on hotplug, enhanced workspace previews in the 'Window List' and 'Workspace Indicator' extensions, and improved handling of non-seekable files in the 'Evince' document viewer.

The Nautilus (files) manager has received several updates, including a file chooser 'choices' menu, drag-and-drop support within the file chooser, vertical scrolling for the pathbar, a new network view, and default listing of physical internal disks in the sidebar. It also features a refreshed sidebar, performance boosts for file search, and support for extracting bzip2 tar archives.